

SH E.656

"Caimano"

SH E.656 Three-Phase Electric Locomotive

Ver. Classic



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1. Introduction

This manual covers the operation of the **SH E.656 'Caimano' Ver. Classic**, an add-on for Train Simulator Classic developed and published by **Silver Hive (SH)**. The Classic edition is the streamlined variant of the locomotive, providing the core controls needed for everyday driving without the advanced procedures found in the Expert version (PAC, SHUNT, IR protection logic).

The model recreates the **SH E.656**, an Italian three-phase electric locomotive in service since 1975, simulating its fundamental on-board systems with authentic fidelity. The RSC-SCMT system is natively integrated by Silver Hive, with no external dependencies.

■ Classic Edition

Available controls: battery, pantographs, spring brake, brake isolating cock, continuous brake, locomotive brake, lights. The RSC-SCMT Silver Hive Simplified system is present and active.

2. Locomotive Overview

2.1 Real Prototype

The **SH E.656 'Caimano'** is based on the Italian E.656 electric locomotive, built in 461 units between 1975 and 1989. Running on **3000 V DC**, it was designed for mixed passenger and freight services. The nickname 'Caimano' (Caiman) derives from the original green livery and the elongated body shape.

2.2 Technical Specifications

Parameter	Value
Type	Electric locomotive
Power supply	3000 V DC
Continuous power	3480 kW (4 motors × 870 kW)
Maximum speed	160 km/h
Service mass	~83 t
Wheel arrangement	Bo'Bo'Bo'
Production period	1975–1989
Units built	461

2.3 Enabling Providers in TSC

Before running a scenario, the Silver Hive provider must be enabled in the Train Simulator Classic editor. The procedure applies to both the locomotive and any support vehicles in the scenario.

Step 1 — Select the SilverHive Provider

In the TSC editor, click the **blue square** (circled in red) in the upper-left corner. From the drop-down menu on the right, select **SilverHive**: the entry **SH_E656_Classic** will appear in the available rolling stock list.



Figure P1 — Selecting the SilverHive provider in the TSC editor

Step 2 — Confirm Selection

Verify that **SH_E656_Classic** is selected in the left panel and confirm its placement in the scenario. No additional providers are required: the RSC-SCMT system is fully integrated within the SilverHive provider.



Figure P2 — SH_E656_Classic selected and ready

■ Note — No External Plugins Required

Unlike the Expert version, the Classic edition requires no third-party plugins. The SilverHive provider alone is sufficient.

3. Cab Overview

3.1 Behind the Driver — Battery

Directly behind the driver's position is the **main battery lever (1)**, which must be engaged before any other on-board system can be operated.



Figure 1 — Behind the driver: battery lever (1)

3.2 Driver's Desk



Figure 2 — Driver's desk with numbered controls (2-11)

No.	Component	Function
2	Pantographs T1 / T2	Raise and lower the two pantographs
3	Reverser	Direction selection: Forward / Neutral / Reverse
4	Notch selector	Traction combinations: 0 M S 0 SP 0 P 0 PP
5	Headlight selector	Front or rear lights (bidirectional)
6	Instrument lights	Dashboard illumination
11	Wheelslip indicators	Wheel adhesion loss warning lights

3.3 Braking System



Figure 3 — Braking system: isolating cock (7), train brake (8), loco brake (9)

No.	Component	Function
7	Brake isolating cock	Isolates or opens the continuous brake circuit
8	Continuous brake (Westinghouse)	Train brake acting on the entire consist
9	Locomotive brake (graduated)	Loco-only brake, fully modulable

3.4 Spring Brake

The **spring brake handwheel (10)** is located behind the driver and serves as the parking brake. It must be fully released before any movement and applied only when the locomotive is stationary at the end of service.

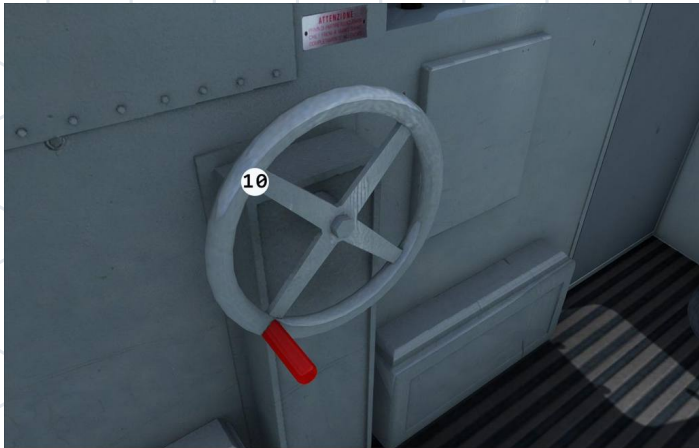


Figure 4 — Spring brake handwheel (10), behind the driver

4. Service Preparation

Follow the steps below in strict sequence. The Classic preparation procedure is simplified compared to the Expert version.

4.1 Battery Activation

- 1

Move to the position behind the driver.
Locate the battery lever (1) on the rear panel.

2

Engage the battery lever (1) — key [B].

Wait approximately 10 seconds: the BAT indicator will flash briefly and then glow steadily. The IR indicator will also illuminate.

■ Caution — Mandatory Wait

Do not proceed until the BAT indicator is steady (not flashing). The on-board systems require approximately 10 seconds to complete initialisation.

4.2 Raising Pantographs

3

Confirm that brakes are applied before raising the pantographs.

The brake isolating cock (7) must be in the open position.

4

Raise pantograph T2 — key [T] or lever (2).

The rear pantograph engages the overhead line. T2 (rear) is preferred when travelling forward.

5

Raise pantograph T1 if required — key [SHIFT+T].

T1 (front) is preferred when travelling in reverse. With both pantographs raised, current collection is redundant.

■ Note — Pantograph Selection

In real-world practice, T2 (rear) is used when running forward and T1 (front) when running in reverse, to prevent the collector strip from dragging against the direction of travel. For normal driving, a single pantograph is sufficient.

4.3 Spring Brake and Cocks

6

Move behind the driver and fully release the spring brake (10) — key [CTRL+Q].

Turn the handwheel anti-clockwise until fully released.

7

Confirm that the brake isolating cock (7) is in the open position.

The continuous brake circuit must be active and unobstructed.

4.4 Starting Traction

8

Set the direction of travel using the reverser (3).

Forward [W] or Reverse [S]. The notch selector (4) is operated with [A] and [D].

9

Move the notch selector (4) to position S (Series) — key [D].

Traction builds progressively. Advance through the notches gradually.

10

Release the continuous brake (8) gradually.

Operate the Westinghouse brake handle to reduce braking pressure.

5. Driving

5.1 Start and Departure

Before moving, verify: BAT indicator steady, at least one pantograph raised, spring brake released, isolating cock open, reverser set to the correct direction.

5.2 Traction — Notch Selector

The notch selector (4) steps through the positions **0 – M – S – 0 – SP – 0 – P – 0 – PP**. Advance gradually to avoid wheel-slip and jerking.

Pos.	Name	Typical Use
M	Shunting	Low speeds, station shunting
S	Series	Starting from rest on the main line
SP	Series-Parallel	Intermediate speed, transition
P	Parallel	Cruising speed
PP	Super-Parallel	High speed on level track

5.3 Braking

To decelerate, return the notch selector to 0 and apply the **continuous brake (8)** via the Westinghouse handle: this brakes the entire consist. The **locomotive brake (9)** is graduated and acts on the locomotive axles only, useful for fine speed adjustments. The **spring brake (10)** must only be applied with the locomotive stationary, as a parking brake.

5.4 Lights

Control	Function
Headlight selector (5) – Fwd	Front headlights on
Headlight selector (5) – Rev	Rear tail lights on
Instrument lights (6)	Dashboard and indicator illumination
Driver's light	Driver reading light
Guard's light	Guard's compartment light

6. RSC-SCMT Silver Hive Simplified System

6.1 Introduction and RSC Codes

The **RSC (Continuous Signal Repetition)** is a safety system that continuously transmits speed codes through the rails. The on-board SSB-SCMT receives these codes and displays them on the desk panel via indicator lights, monitoring compliance with the authorised speed limits.

The **RSC-SCMT Silver Hive Simplified** system is natively integrated and requires no third-party plugins. It replicates the core logic of the real system, including speed codes, the PRE and RIC buttons, and automatic emergency braking.

RSC Codes and Indicator Lights

RSC codes are received from the track and indicate the state of the next signal ahead. Each code corresponds to a specific colour on the desk panel:

Code	Icon	Meaning	Speed limit
75	▲ yellow/ black triangle	Next signal at Danger within 900 m or more. Reduce speed in accordance with the braking curve.	V1
120	RV yellow	Next signal clear for a diverging route. Maximum speed 30 or 60 km/h depending on the signal aspect passed.	V60

180	■ white square	Clear with advance warning. The next signal is clear but a speed reduction is expected further ahead.	Vr
270	■ green square	Clear for at least 2700 m ahead. No speed restriction expected in the following section.	V2
AC	AC icon (grey)	No RSC code received. The line is not equipped with RSC or the system is awaiting activation.	—



Figure 5 — RSC panel with AC indicator lit (system awaiting activation)

6.2 RSC Activation

At the start of service the desk shows the **AC** indicator lit: the RSC system is powered but not yet manually engaged. Press the **RSC button** to activate it.

11

Press the RSC button — key [CTRL+R].

The RSC indicator on the desk illuminates with a steady **blue light**, confirming that RSC is engaged. The system begins receiving codes from the track.

■ **Note — No Code on Unequipped Lines**

On sections without RSC (lines not fitted with BAcc track circuits) the AC indicator will remain lit. The system stays active and will update automatically upon receiving the first code.

6.3 PRE Button — Pre-Acknowledgement

The **PRE** button illuminates when the SSB-SCMT detects an event requiring driver acknowledgement: approaching a caution signal, a set of points, or a restrictive RSC code change. The driver must press PRE within **3 seconds** of the light coming on.



Figure 7 — PRE button lit: press within 3 seconds

12

Press PRE as soon as the button illuminates — key [CTRL+P].

The PRE button goes out, confirming receipt. The RIC button illuminates immediately after.

■ Caution — Automatic Emergency Braking

If PRE is not pressed within 3 seconds of illumination, the SSB-SCMT triggers automatic emergency braking (RF). Always keep attention on the RSC panel while under way.

6.4 RIC Button — Acknowledgement

The **RIC** button illuminates immediately after PRE has been pressed. The driver must press RIC **before passing the first relevant signal**. After pressing RIC, the system allows **20 seconds** to clear the signal; if that window expires without passage, emergency braking is triggered.



Figure 6 — RIC button lit: press before passing the signal

13

Press RIC before passing the signal — key [CTRL+K].

After pressing, 20 seconds are available to clear the signal. The system resets to normal operation once the signal is passed.

■ RSC Sequence Summary

RSC active → code received → PRE lights up → press PRE within 3 s → RIC lights up → press RIC before signal → clear signal within 20 s → system reset.

7. Keys and Controls

The table below lists the default key bindings for the SH E.656 Ver. Classic. Bindings can be customised through the Train Simulator Classic interface.

Key	Function	Key	Function
W / S	Reverser: Forward / Reverse	H	Headlights — front / rear
A / D	Notch selector: step down / up	SHIFT+H	Instrument lights (6)
T	Pantograph T2 Up/Down	CTRL+R	Engage/disengage RSC
SHIFT+T	Pantograph T1 Up/Down	CTRL+P	PRE button — pre-acknowledgement
,	Continuous brake — apply	CTRL+K	RIC button — acknowledgement
.	Continuous brake — release	B	Battery On/Off
[	Locomotive brake — apply	F1	Internal cab view
]	Locomotive brake — release	F2	External view
CTRL+Q	Spring brake — apply/release	SHIFT+F1	View behind driver
BACKSPACE	Emergency brake		

8. Shutdown

Deactivation procedure at the end of service:

14	Return the notch selector (4) to position 0 — key [A]. Confirm that traction is fully disengaged.
15	Return the reverser (3) to Neutral. Centre position.
16	Apply the spring brake (10) — key [CTRL+Q]. Move behind the driver and turn the handwheel clockwise until fully locked.
17	Isolate the continuous brake using the cock (7). Set to the isolated position to cut off the pneumatic circuit.
18	Lower pantographs T2 and T1. Press [T] then [SHIFT+T]. Confirm both are fully lowered.
19	Disengage the battery (1) — key [B]. All indicators go out. The locomotive is deactivated.

9. Troubleshooting

Symptom	Likely Cause	Solution
BAT indicator flashing for a long time	Initialisation in progress	Wait 10 seconds after engaging the battery
Pantographs will not raise	Battery not engaged or not yet initialised	Wait for BAT to glow steadily, then raise pantographs

Symptom	Likely Cause	Solution
Traction unresponsive	Spring brake not released	Move behind the driver and release the handwheel
RSC triggers emergency braking	PRE not pressed within 3 s, or RIC not pressed before signal	Set notch to 0, wait, then re-engage RSC with [CTRL+R]
AC indicator permanently lit	RSC not activated or line not equipped with BAcc	Press [CTRL+R] to engage RSC
Loco will not move after shutdown	Preparation sequence not followed correctly	Repeat the full preparation procedure from step 1

10. Credits and Licence

Silver Hive (SH) — Development and Publishing

The **SH E.656 'Caimano' Ver. Classic** is a commercial product developed and published by **Silver Hive (SH)**. The 3D model, textures, simulation code, integrated RSC-SCMT system, and this manual are the exclusive work of Silver Hive.

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■ Support

For bug reports, updates, and support, refer to the official Silver Hive channels. This manual is subject to revision following new releases of the add-on.